Clara Hoffmann

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Education

PhD candidate in Computer Science - Karlsruhe Institute of Technology 02/2023 - current
Methods for Big Data at Scientific Computing Center (SCC) & DesBi *Karlsruhe & Dortmund, Germany*
Developing Bayesian Deep Learning methods for active sensor time-series data, supervised by Prof. Dr. Nadja Klein. Initially at TU Dortmund and continued at KIT following my supervisor's chair relocation.

M.Sc. in Statistics - Humboldt University 10/2018 – 07/2021
Bayesian statistics, machine and deep learning. GPA: 1.2/1 (German scale, 1.0 = best) *Berlin, Germany*
Master's thesis: Marginally calibrated response densities for end-to-end (EtE) learning in autonomous driving (1.0).

B.Sc. in Economics - Humboldt University 10/2013 – 02/2017
Semester abroad at Maastricht University, Netherlands. GPA: 1.7/1 *Berlin, Germany*

Academic & Industry Experience

Machine Learning Engineer - Change Detection 11/2021 – 12/2022
LiveEO - Satellite Based Infrastructure Monitoring *Berlin, Germany*
ML-based change detection for small objects using SAR satellite imagery (ICEYE, Capella). Prototyping models and pre-processing SAR imagery, setting up scalable imagery workflows, and delivering near-real-time analyses.

Student Research Assistant - Bayesian Statistics 06/2021 – 10/2021
Chair of Applied Statistics - Humboldt University *Berlin, Germany*
Research project on copula-based marginally calibrated regression for discrete responses with spatially structured selection priors for Prof. Dr. Nadja Klein. Deriving the model structure and implementing the model.

Publications & Talks

Bayesian meta-learning for modeling Alzheimer's disease progression | *arXiv preprint* 06/2026
Identifying an overconfidence problem in single-task models for predicting Alzheimer's disease progression from short time-series, and addressing it with a partially Bayesian meta-learning model. Authors: Clara Hoffmann & Nadja Klein.

Marginally calibrated response distributions for EtE-learning in autonomous driving | *AoAS* 06/2023
Scalable estimation of predictive densities for quantifying the uncertainty of steering angles in end-to-end learners using a deep implicit copula. Authors: Clara Hoffmann & Nadja Klein, published in *Annals of Applied Statistics*.

Honey, I broke the PyTorch model | Talk at PyCon & PyData Berlin 04/2023
Building a toolbox of synthetic data and tests to identify bugs in model training (watch the full talk on YouTube).

Marginally Calibrated Response Densities for EtE Learning | Talk at *Statistical Week 2022 & SEIO 2023* 09/2022
Presentation about scalable and reliable uncertainty quantification of end-to-end learners (animated slides available here).

Awards

Mitchell Prize | *International Society for Bayesian Analysis (ISBA)* 2024
Awarded to "*Marginally calibrated response densities for EtE learning in autonomous driving*" as an outstanding paper demonstrating how Bayesian analysis solved an important applied problem.

Young Statistician Award for the best Master's thesis | German Statistical Association (DStatG) 2022

Technical Skills & Languages

Programming: Python (PyTorch, JAX, NumPy, pandas, rasterio, geopandas), unit testing, typing, SQL, shell scripting, experience with handling GB/TB datasets
Tools & Platforms: AWS (S3, EC2, Batch), Git, uv, poetry, Docker **Organizational**: Jira, Confluence, Scrum
Spoken Languages: German (native), English (fluent), Mandarin (intermediate), Spanish (intermediate)